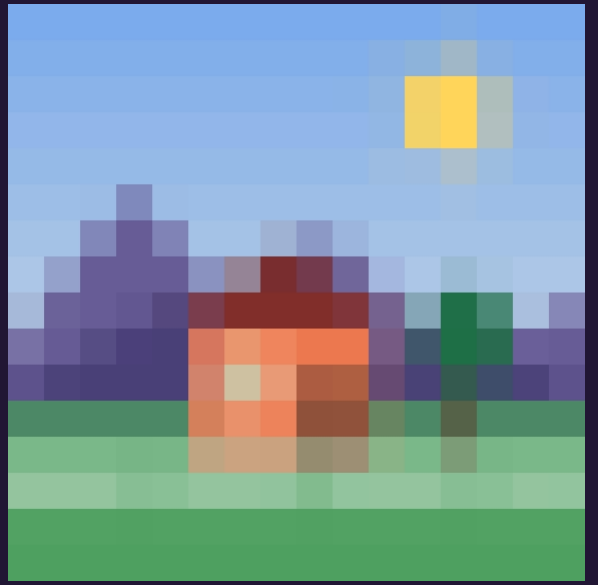
Spatial Resolution in Image Processing

From pixels and sampling to aliasing and interpolation

Digital Image Processing | Technical Presentation

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The same scene at 16 × 16 pixels

Abstract

Spatial resolution is the number of pixels that sample a scene, and so the amount of fine detail an image can hold. This talk explains how images are sampled, what is lost as the pixel count falls, why coarse sampling creates false patterns (aliasing), and how interpolation enlarges an image without restoring detail. Worked examples on one test scene show the trade-off between detail and cost.

Sampling | Aliasing | Nyquist | Interpolation



Test scene used in every example

Introduction

SAMPLING

Spatial resolution

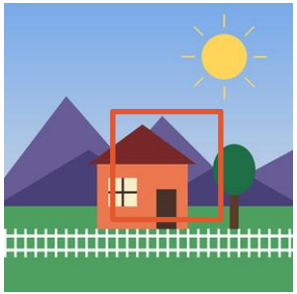
How many pixels (M × N) sample the scene.

QUANTIZATION

Intensity resolution

How many gray levels: 2^k for k bits (8 bits = 256 levels).

Measured in pixels (M × N), megapixels, PPI/DPI or line pairs per mm.



Zoom into a patch: every pixel is one number from 0 (black) to 255 (white).

189	176	154	178	189	189
139	68	77	111	182	194
71	71	71	76	108	158
170	160	150	150	103	82
193	173	114	114	86	73
167	158	97	98	121	114

6 × 6 patch = 36 numbers

Objectives

1. Explain how a scene is sampled into pixels and gray levels.
2. Show how reducing the pixel count removes detail and saves storage.
3. Demonstrate aliasing and how filtering before sampling prevents it.
4. Compare interpolation methods for enlarging an image.

Problem Statement

THE QUESTION

How many pixels are enough?

The answer depends on the task: medical, satellite, document and surveillance imaging all choose differently.

Too few pixels

Fine detail such as thin lines and small text is lost for good.

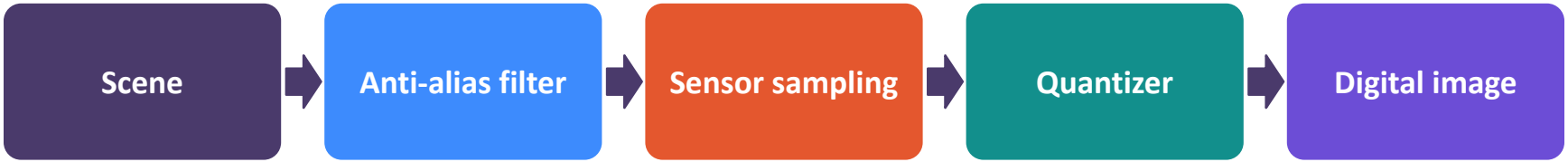
Too many pixels

Each doubling of width needs 4× the storage, bandwidth and compute.

Coarse sampling, no filter

Detail turns into false patterns (aliasing) that were never in the scene.

Architecture

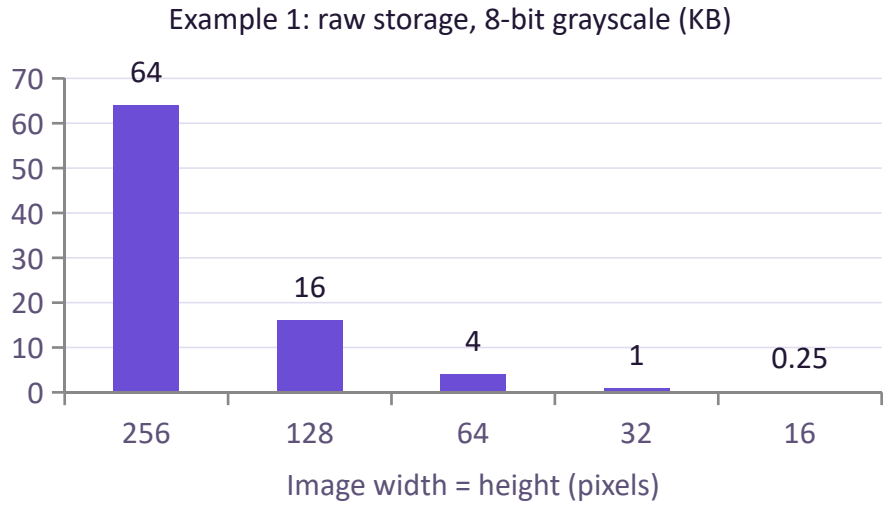


Scene	Anti-alias filter	Sensor sampling	Quantizer	Digital image
Continuous light with unlimited detail.	Blurs detail finer than the pixel grid can hold.	Picks M × N points. This sets spatial resolution.	Rounds each value to one of 2^k gray levels.	An array of integers f(x, y).

Later steps (resize, interpolate, display) cannot add detail: resolution is fixed at the sampling stage.

How It Works: Shrinking Pixels

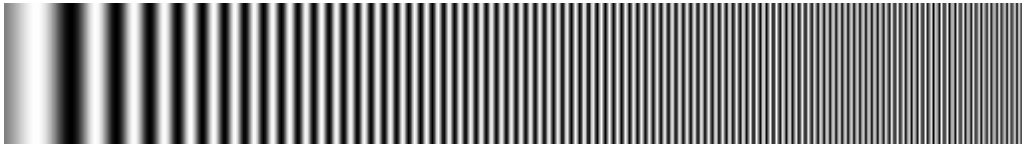




Halve the width and you keep only ¼ of the pixels, ¼ of the storage.

How It Works: Aliasing

Example 2: original pattern, stripes get finer to the right



Naive sampling every 8th pixel: false wide stripes appear



Blur first, then sample: detail fades, no false pattern



NYQUIST RULE

$$f_s \geq 2 \cdot f_{max}$$

Take at least 2 samples per cycle of the finest detail.

Below that, fine detail folds into false coarse patterns (moiré).

Fix: low-pass filter before downsampling.

How It Works: Interpolation



Nearest neighbour

Copies the closest pixel: blocky but fastest.



Bilinear

Averages 4 neighbours: smooth but slightly soft.



Bicubic

Fits 16 neighbours: sharper edges, mild ringing.


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The same scene at 16 × 16 pixels
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+9hp, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.025906° Long 80.016379°
Monday, 26/10/2026 01:07 PM GMT +05:30
Google

Conclusion

1

More pixels, more detail

But every doubling of width costs 4× the data.

2

Sample fast enough

At least 2 samples per feature, and filter before shrinking.

3

Interpolate for looks, not truth

Zooming smooths blocks but cannot restore lost detail.

Choose the lowest resolution that still meets the task. Thank you! Questions?